Pranav Puttagunta

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Education

University of California San Diego, *Jacobs School of Engineering* — Expected 2027
Bachelor of Engineering in Computer Science — GPA: 3.92/4.0

Technical Skills

Languages: C++, Python, C, MATLAB, Bash, SQL, Java, Assembly, TypeScript
Robotics/Sim: ROS/ROS2, Gazebo, Webots, URDF/SDF, MAVLink, PX4/Pixhawk, SLAM, Motion Planning (MCTS)
AI & Perception: OpenCV, Point Clouds, CUDA, TensorFlow, PyTorch, MediaPipe, NumPy, Sensor Fusion
Hardware/Tools: Linux (Ubuntu), Docker, Git, SolidWorks, Altium, Arduino, Raspberry Pi, LiDAR/GPS/IMU

Experience

Yonder Dynamics, Autonomous Systems Developer – La Jolla, CA — Oct. 2024 – Present
- Enabled validation of autonomous logic by designing physics-based SITL simulations with URDFs, saving 50 hours.
- Raised precision from 10m to 10cm by writing ROS nodes to fuse RTK, Pixhawk, and RM3100 compass via MAVROS.
- Reduced mission failure rates by 30% by designing autonomous return-to-base logic triggered by signal loss.
- Ensured consistent ROS integration across machines with Bash scripts to automate Docker environment configuration.

Brains4Drones, Software Engineering Intern – Plano, TX — Mar. 2022 – Dec. 2024
- Led development of PreCheck LiDAR tool, cutting drone failures by **60%** with pre-flight terrain modeling and analysis.
- Optimized runway maintenance with TensorFlow crack detection pipelines, cutting manual inspection time by **50%**.
- Mitigated collision risks by creating LiDAR flight simulations, allowing for safe validation of complex mission paths.
- Accelerated point-cloud obstacle detection for drone flight by designing GPU-optimized CUDA pipelines with KNN.

Pragma Edge (IBM Partner), AI Engineering Intern – Jacksonville, FL — Oct. 2025 – Present
- Addressed inefficiencies by architecting REST APIs for IBM Maximo, enabling automated inventory management.
- Integrating APIs into IBM WatsonX chatbots with LLMS to enable natural language querying of large asset datasets.
- Developing AI computer vision (OpenCV, TensorFlow) pipelines to automate defect analysis and asset monitoring.
- Integrating MLOps pipelines in Python (OpenCV, TensorFlow) for scalable enterprise data flows for anomaly detection.

UCSD Advanced Robotics and Control Lab, Research Assistant – La Jolla, CA — Mar. 2025 – Oct. 2025
- Automated wound analysis by developing 3D mesh reconstruction algorithms (Open3D/SDFs) with **80%** accuracy.
- Validated clinical feasibility by collaborating with PhD students to test algorithms on humanoid robot prototypes.
- Enabled gauze application by implementing MCTS + heuristics motion planning with **100%** wound coverage.
- Optimized algorithmic efficiency, reducing runtime by **200%** compared to baselines without sacrificing performance.

Wheelhouse Robotics, FIRST/VEX Robotics Instructor – Coppell, TX — Jun. 2025 – Sep. 2025
- Guided FRC team to prototype a swerve robot in **1** week and deploy vision-based autonomous navigation in **2** weeks.
- Coached **28** students across **4** teams, teaching CAD, Java, Python, Git workflows, and Computer Vision principles.
- Improved technical collaboration and design reviews by organizing PDRs, debugging sessions, and Git workflows.

Projects & Portfolio Highlights

VisLink | ***Python, OpenCV, MediaPipe*** — **Team Lead**
- Architected a CV/ML-driven desktop navigation interface for paralyzed users, winning 1st place at UCD Hackathon.
- Engineered a low-latency vision pipeline with OpenCV + Google MediaPipe achieving **80%** accuracy.
- Enhanced cursor stability and tool usability by integrating signal smoothing algorithms and voice recognition libraries.

SideKick | ***React Native, FastAPI, Redis, AWS, Docker*** — **Lead Architect**
- Analyzed user biomechanics by developing an OpenCV + MediaPipe pipeline to extract form metrics for LLM analysis.
- Scaled video processing capabilities by building an AI coaching app on AWS using React Native and Redis + Celery.
- Secured user data and delivered real-time insights by building a REST API with Firebase authentication.

OpenLabel | ***Python, OpenCV, Gemini API, Tkinter*** — **Lead Developer**
- Architected a vision/AI based allergen/diet recommendation app, winning Robin Hood track at UCSD Hackathon.
- Parsed ingredient labels in real-time by architecting an image processing pipeline utilizing the Gemini API + CV.
- Delivered personalized purchasing advice by implementing LLM logic tailored to user health restrictions.

Honors and Leadership

SacHacks 1st Place • DiamondHacks 1st Place • NASA Moonshot System Lead • FIRST Robotics Team Founder • PURE Nonprofit Chapter Director • Presidential Gold Service Award • National Merit Finalist • Taekwondo National Medalist